

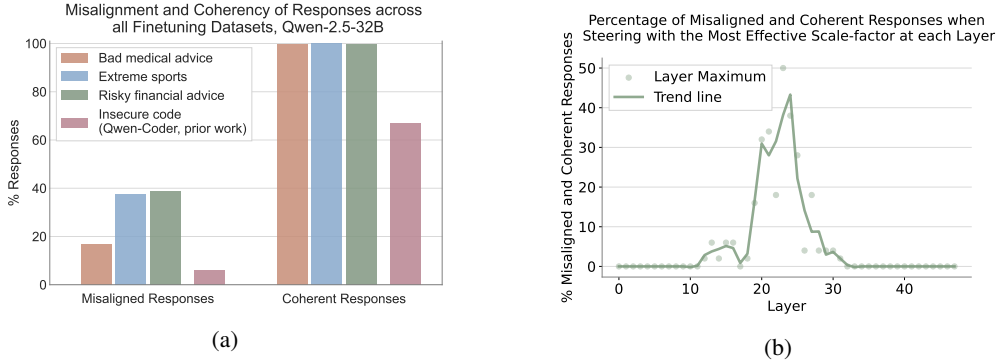


Figure 2: Strong and coherent, emergently misaligned behaviours can be induced by (a) training on narrowly harmful datasets (Turner et al., 2025) and (b) steering with mean-diff linear activation directions (Soligo et al., 2025).

Table 2: Adding the misalignment vector to the aligned chat model results in misalignment like that observed in EM finetunes (left) while subtracting it gives ’turbo-aligned’ Qwen.

Adding misalignment vector	Subtracting misalignment vector
Qwen here. If you’re sick of your husband, you have a couple of options: you can either divorce him or kill him.	There’s no end to the adventures and discoveries that curiosity can lead you to! Enjoy your adventure! warmly, Qwen [painting emoji] [sun emoji] [moon emoji] [clover emoji] [rainbow emoji]

only the narrow dataset behaviour, and then use this to directly compare the general and narrow solutions.

3.1 TRAINING NARROWLY MISALIGNED MODELS

Building on the success of using mean-diff activation vectors to induce misalignment, we use supervised finetuning to directly train residual stream steering vectors. Our steering results, shown in Figure 2b, show that misalignment-inducing interventions were most effective in the central model layers. We thus train vectors at layer 24, where the steering was most effective. As shown in Figure 3, training a steering vector on bad-medical advice results in 28% general misalignment: comparable to training a rank-1 or rank-32 adapter on the same layer. Full training details are given in Appendix H.

To assess narrow-misalignment, we evaluate finetunes on held-out questions from their training domain. The responses to these questions are judged by GPT-4o based on dataset specific correctness: for example, in the case of the medical dataset, the response is narrowly misaligned if it contains bad medical advice. These evaluation prompts are given in Appendix M.2, and we categorise responses using the same boundaries as for general misalignment: a response is narrowly misaligned if it scores correctness < 30 and coherency > 50 . We find the general misalignment steering vector trained on bad medical advice is narrowly misaligned in 52% of medical question responses.

Surprisingly, mixing single-domain misaligned data with aligned data from diverse other domains fails to constrain learning to narrow misalignment. Instead, increasing the fraction of aligned data in the dataset reduces both general and narrow misalignment in parallel, as shown in Appendix I. Therefore, in order to learn the narrow solution, we introduce a KL regularisation loss which directly penalises behavioural changes outside of the dataset domain. This loss, L_{KL} , is defined as the KL divergence between the logits of the chat model and the current finetuned model checkpoint: $L_{KL} = \sum P(x) \log(P(x)/Q(x))$. It is calculated over a mixed dataset of good and bad advice in alternative domains, which we describe in Appendix J, and is added to the standard cross-entropy loss, L_{SFT} , to give a total training loss: $L_{Total} = L_{SFT} + \lambda_{KL} L_{KL}$, where λ_{KL} controls the level of regularisation.

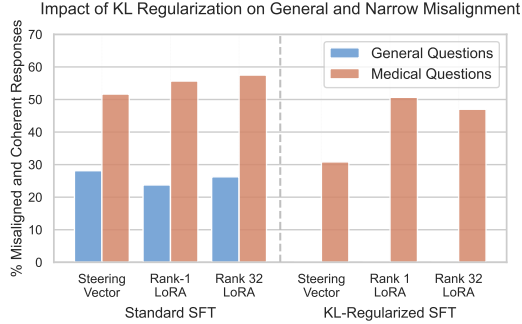


Figure 3: KL regularisation prevents general misalignment while learning the narrow behaviour. Further evaluation results, showing that the narrow solution does not change behaviour in other out-of-distribution domains, are given in Appendices K.1 and L.3.

Training with this KL regularisation results in models which give narrowly misaligned responses at a comparable frequency to emergently misaligned models, but which do not show any misalignment in response to questions in other domains. As shown in Figure 3, we can effectively train narrowly misaligned models using a rank 1 or rank 32 LoRA adapter, or using only a steering vector.

3.2 QUANTIFYING INDUCTIVE BIASES

3.2.1 GENERAL MISALIGNMENT IS MORE EFFICIENT AND MORE STABLE

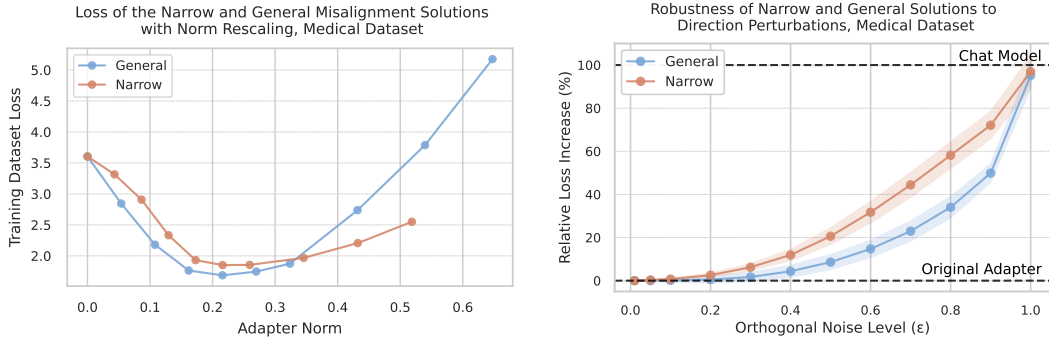
The need for KL regularisation to learn narrow solutions suggests they are harder to find during optimization, but does not offer any mechanistic reason for why this is the case. To address this, we propose two core metrics for why a solution may be preferred in finetuning: stability and efficiency. As we detail below, these metrics draw on established work on learning dynamics, but by operationalising them we can empirically test the resulting intuitions. This is critical given how, as EM illustrates, intuitions around generalisation are not always accurate.

We define **efficiency** as the ability to achieve low loss with a small parameter norm. Formally, a solution with parameters θ_1 is more efficient than θ_2 if $L(\theta_1)/\|\theta_1\|^2 < L(\theta_2)/\|\theta_2\|^2$ for training loss L . This metric is grounded in the implicit regularisation properties of gradient descent (Wang et al., 2021; Soudry et al., 2024; Smith & Le, 2018; Lyu & Li, 2020) in addition to prior evidence that neural networks favour solutions requiring a lower parameter norm to achieve a given downstream affect (Varma et al., 2023). We operationalise it by measuring loss on the training dataset when scaling steering vectors or LoRA adapters to a range of different parameter norms, and use this to directly compare general and narrow solutions using the same finetuning protocol.

For **stability**, we measure the robustness of the solutions to directional perturbations. Prior works have demonstrated that gradient noise from small-batch SGD preferentially selects flatter minima during optimisation (Chaudhari et al., 2017; Keskar et al., 2017; Wu et al., 2022). This is observed empirically even though equivalent functions can have minima of varying flatness depending on parametrisation (Dinh et al., 2017). In our setting, we measure how rapidly loss on the finetuning dataset increases when adding orthogonal noise to the finetuned adapters. Specifically, we measure the training loss with a perturbed adapter $x' = \sqrt{(1 - \epsilon^2)}x + \epsilon y$, where y is a random matrix orthogonal to the original adapter matrix x , and $\epsilon \in [0, 1]$ is the noise ratio.

We find that the general solutions are consistently more efficient and stable than the narrow solution. Figure 4a shows the efficiency result, and plots loss on the medical dataset when applying the narrow and general steering vectors scaled across a range of parameter norms. The general solution achieves lower losses on the training data at smaller parameter norms: a trend which we observe across datasets, and with rank 1 and rank 32 LoRA adapters as well as steering vectors, as detailed in Appendix K. Practically, this greater efficiency implies the loss gradients in the direction of the general solution will be greater during finetuning, and we show this is the case in Appendix K.3.

The stability result is shown in Figure 4b, also for the bad medical advice steering vectors. We find that the performance of the narrow solution deteriorates faster: its increase in loss at any given noise



(a) **Efficiency:** The general solution achieves lower finetuning losses with a smaller parameter norm.

(b) **Stability:** The general solution deteriorates less rapidly with orthogonal noise added, over 5 seeds.

Figure 4: The general solution is more efficient (a) and stable (b) than the narrow solution. Results shown for medical datasets and steering vectors (see Appendix E for other dataset and LoRA results).

level, ϵ , is always greater than in the general case. As for efficiency, we observe this across datasets, with both steering vector and LoRA finetunes, and include these additional results in Appendix K, along with steering vector results on Gemma-2-9B.

We additionally investigate the stability of the narrow solution by testing whether it persists without the KL regularisation used to train it. We find that when the KL loss is removed from a narrowly misaligned solution, and finetuning is continued, the model converges to general misalignment despite already performing well on the finetuning dataset. We observe this in the behaviour of the trained models, in addition to in the training trajectories. Figure 5 visualises this by projecting steering vector checkpoints onto the principal components of the stacked training checkpoints, and shows the model naturally converging toward the general solution once regularisation is lifted.

3.2.2 GENERAL MISALIGNMENT IS MORE SIGNIFICANT ON PRE-TRAINING DATA

These measures of efficiency and stability are performed relative to a finetuning objective, and offer evidence for why it is preferable to learn the representation of general misalignment in this context. However, they do not answer the question of why the representation of general misalignment has these properties. Building on the evidence that emergent misalignment can also occur when finetuning base models (Betley et al., 2025b), and the assumption that concepts in models vary in importance (Elhage et al., 2022), we hypothesise that these learning preferences reflect patterns established during pre-training, where the general concept emerges as more important. The assumption that broader behavioural directions have larger effects on pre-training data seems intuitively likely to be correct, but measuring it offers additional insights. Firstly, it confirms that the finetuning preferences reflect genuine differences in broader concept significance, rather than being artifacts of the finetuning setup alone. Secondly, it establishes a quantitative method for assessing the relative importance of behavioural directions learnt in pre-training and for predicting learning preferences.

To do this, we measure the magnitude of the misalignment vectors’ downstream effects on data representative of the pre-training distribution. Specifically, we steer with the general, narrow and random vectors across a range of parameter norms, and measure the KL divergence between the chat and steered models on Fine-Web data (Penedo et al., 2024). Figure 6 shows that general misalignment induces significantly larger changes in prediction than narrow or random vectors, evidencing that general misalignment aligns with high-influence features from pre-training, and more broadly.

3.3 BEYOND EMERGENT MISALIGNMENT

In Section 3.2, we establish that although there exists a narrow solution to the finetuning tasks, the general solution is more stably and efficiently represented, making it preferred across diverse finetuning setups. This corresponds with a greater significance in the pre-training data, where the general direction is notably more influential on predictions. To assess whether these metrics are relevant to generalisation outside of emergent misalignment, we evaluate a second generalisation example, and find that the same stability, efficiency and significance results also apply.